Failure Analysis Report

FAIL-0782 — Data Mismatch

1. Failure Summary

Failure ID	FAIL-0782
Failure Type	Data Mismatch
Severity	Critical
Component	Firmware Manager
Application	APP-001 — SmartHome
Build Version	4.5.0
Timestamp	2023-02-07 04:25:22
Environment	ENV-009 — Low Network Env
OS Version	Android 14
Network	Low-BW Sim
Resolution	Open

2. Test Case Reference

Test Case ID	TC-1541
Title	Verify that push notification delivery completes successfully when the input dat...
Execution ID	EXEC-04076
Test Run	RUN-010 — Regression Cycle 10 — Build 6.2.3
Requirement	REQ-0192 — Battery — Background Sync

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete media streaming within the defined SLA.
Observed Behaviour	The system exhibited data mismatch when the session token has expired.

4. Error Details

Error Message:

```
DataIntegrityError: field 'token' expected=true actual=TIMEOUT.
```

Stack Trace / Log Excerpt:

```
com.samsung.health.Database.execute(line 725)
```

Log Reference	LOG-68478

5. Executed Steps at Time of Failure

1	Launch the VoiceAssistant application on the test device.	A success toast message is displayed.

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2	Navigate to Settings > Privacy and tap 'Pair Device'.	The device reconnects automatically within 10 seconds.
3	Disconnect the wearable and wait 9 seconds.	A success toast message is displayed.

6. Root Cause Analysis

Missing null-check on the device registry response before dereferencing the session token.

7. Linked Defect

Defect ID	BUG-0220
Title	Data Mismatch in Firmware Manager: DataIntegrityError: field 'token' expected=tr...
Severity	Critical
Priority	Low
Status	OPEN
Root Cause	Missing null-check on the device registry response before dereferencing the sess...
Resolution	—
Assigned To	ENG-074

8. Historical Similar Failures

HIST-0156	FAIL-0157	Data Mismatch	3.1.1	2021-03-21	3	Open
HIST-0398	FAIL-0399	Data Mismatch	6.2.8	2023-09-18	2	Wont Fix
HIST-0445	FAIL-0446	Data Mismatch	6.0.2	2023-02-14	1	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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